

ZARAGOZA AP, Spain

WMO: 081600

Lat: 41.6606N

Lon: 1.0042W

Elev: 258

StdP: 98.26

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-2.0	-0.3	-8.2	1.9	4.3	-5.9	2.4	4.0	14.3	10.4	13.1	9.0	2.5	290	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	36.6	20.8	34.9	20.4	33.1	19.9	22.2	32.2	21.5	31.1	20.8	30.2	4.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.6	24.7	18.5	13.8	24.6	17.8	13.1	24.5	66.3	32.3	63.6	31.3	61.2	30.4	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.1	11.8	10.7	DB	-3.9	39.7	1.8	1.5	-5.1	40.8	-6.2	41.7	-7.1	42.5	-8.4	43.6	(4)
(5)			WB	-4.8	24.0	1.8	0.9	-6.1	24.7	-7.2	25.2	-8.2	25.6	-9.6	26.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	7.2	8.4	11.6	14.2	18.3	23.0	25.4	25.2	21.2	16.8	10.7	7.1	(6)
(7)		DBStd	7.32	3.18	3.09	3.09	3.17	3.49	3.62	2.98	2.84	3.00	3.16	3.20	3.53	(7)
(8)		HDD10.0	311	97	61	19	4	0	0	0	0	0	1	28	101	(8)
(9)		HDD18.3	1660	346	278	208	127	44	5	1	0	8	67	228	349	(9)
(10)		CDD10.0	2433	10	16	70	131	257	390	478	471	336	213	51	11	(10)
(11)		CDD18.3	738	0	0	1	4	43	144	220	213	93	21	0	0	(11)
(12)		CDH23.3	7308	0	0	7	66	431	1506	2334	2115	728	121	1	0	(12)
(13)		CDH26.7	3366	0	0	0	9	134	721	1203	1038	243	18	0	0	(13)
(14)	Wind	WSAvg	4.6	4.6	5.1	4.9	4.9	4.8	4.8	5.0	4.7	4.2	3.8	4.5	4.0	(14)
(15)	Precipitation	PrecAvg	316	23	22	24	36	36	30	15	17	25	35	34	20	(15)
(16)		PrecMax	542	81	70	63	127	142	100	58	88	101	116	104	77	(16)
(17)		PrecMin	186	0	0	0	0	1	0	0	0	0	0	0	0	(17)
(18)		PrecStd	77	20	15	18	29	27	25	14	19	25	29	27	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	19.2	24.1	28.0	33.0	38.1	38.3	38.2	34.1	28.9	21.2	17.8	(19)
(20)			MCWB	11.8	11.8	13.5	15.7	18.7	20.6	21.2	21.2	20.2	17.8	14.4	12.2	(20)
(21)		2%	DB	15.2	17.2	21.9	25.1	30.0	35.2	36.6	36.2	31.3	26.3	18.9	15.2	(21)
(22)			MCWB	10.6	10.7	12.6	14.7	17.4	19.9	21.0	20.7	19.2	17.0	13.4	10.8	(22)
(23)		5%	DB	13.9	15.8	19.9	23.0	27.9	33.1	34.9	34.1	29.3	24.3	17.1	13.8	(23)
(24)			MCWB	9.6	9.9	11.9	13.8	16.6	19.4	20.6	20.4	18.6	16.3	12.1	9.9	(24)
(25)		10%	DB	12.1	14.0	17.9	21.0	25.8	31.0	33.0	32.2	27.6	22.3	15.8	12.1	(25)
(26)			MCWB	8.4	9.1	11.1	13.0	15.8	18.7	20.1	20.0	17.9	15.4	11.5	9.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	12.9	14.7	16.6	19.6	22.5	23.3	23.3	21.7	19.5	15.6	12.8	(27)
(28)			MCDWB	16.5	17.6	21.9	25.2	30.3	34.1	34.6	33.6	29.7	25.1	19.6	16.2	(28)
(29)		2%	WB	11.1	11.4	13.5	15.4	18.2	21.1	22.2	22.3	20.7	18.4	14.2	11.3	(29)
(30)			MCDWB	14.7	16.0	19.9	23.4	27.5	31.9	32.9	31.6	28.2	23.7	17.3	14.3	(30)
(31)		5%	WB	10.1	10.3	12.5	14.5	17.3	20.2	21.5	21.6	19.8	17.5	13.1	10.4	(31)
(32)			MCDWB	12.8	14.5	18.5	21.6	25.6	30.6	31.8	30.5	26.5	22.1	15.8	13.0	(32)
(33)		10%	WB	9.1	9.5	11.5	13.5	16.4	19.4	20.8	20.8	18.9	16.6	12.2	9.5	(33)
(34)			MCDWB	11.6	13.3	16.9	19.6	23.9	28.9	30.7	29.6	25.5	20.8	14.8	11.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.6	9.3	11.0	11.2	12.1	13.5	13.9	13.1	11.5	10.1	8.0	7.4	(35)
(36)			MCDBR	9.5	11.9	14.4	14.7	15.5	16.5	16.3	15.7	13.9	12.5	9.6	9.0	(36)
(37)			MCWBR	6.0	6.7	7.4	6.9	6.4	6.2	6.0	5.7	5.6	5.6	5.5	5.7	(37)
(38)		5% WB	MCDWB	8.3	10.5	12.7	13.3	13.9	15.3	15.2	14.2	12.4	10.7	8.2	8.1	(38)
(39)			MCWBR	5.7	6.3	6.9	6.7	6.2	6.2	6.0	5.9	5.3	5.2	5.3	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.299	0.313	0.354	0.375	0.390	0.400	0.388	0.393	0.373	0.350	0.316	0.298	(40)	
(41)		taud	2.492	2.451	2.354	2.311	2.302	2.302	2.326	2.315	2.371	2.443	2.490	2.499	(41)	
(42)		Ebn at Noon	859	898	891	895	886	874	881	866	860	841	827	827	(42)	
(43)		Edh at Noon	74	89	111	124	128	129	125	122	108	89	73	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.97	2.92	4.17	5.28	6.34	7.13	7.35	6.37	4.92	3.34	2.19	1.73	(44)	
(45)		RadStd	0.19	0.31	0.41	0.44	0.39	0.26	0.22	0.23	0.20	0.25	0.21	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.93	N/A	N/A	N/A	N/A	+102	+92
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page